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Received: 14 December 2025

Accepted: 9 April 2026

Cite this article as: Luo, M., Liu, Z., Huang, Q. *et al.* Uniform and ordered self-assembled hole-selective layers driven by π - π interactions for efficient perovskite-silicon tandem solar cells. *Nat Commun* (2026). <https://doi.org/10.1038/s41467-026-72160-x>

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Uniform and Ordered Self-Assembled Hole-Selective Layers Driven by π - π Interactions for Efficient Perovskite-Silicon Tandem Solar Cells

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Abstract

Self-assembled molecules (SAMs) are promising hole-selective layers for high-performance perovskite tandem solar cells. However, their inhomogeneous distribution and disordered packing on substrates lead to interfacial energy losses, limiting further improvements in efficiency and stability. Here, we design a SAM, Me-Ph2mPACz, through *meta*-disubstitution of dimethylcarbazole moieties on a phenyl linker. Compared to its monosubstituted carbazole counterpart, Me-PhpPACz, Me-Ph2mPACz exhibits stronger adsorption energy and suppresses intermolecular hydrogen bond interaction via π - π stacking interactions. This inhibits the formation of large micelles, promoting a uniform, ordered and thermoresistant hole-selective layer. The resulting multilayer configuration retards crystallization and alleviates residual stress in the perovskite film, thereby reducing non-radiative recombination at the buried interface and enhancing hole extraction. The implementation of Me-Ph2mPACz as the hole-selective layer in 1.68 eV perovskite solar cells reduces interfacial non-radiative losses from 168 mV to 124 mV, accompanied by an increase in power conversion efficiency from 21.82% to 23.14%. The corresponding perovskite-silicon tandem solar cells achieve a champion PCE of 33.40% (certified 32.45% at National Renewable Energy Laboratory,

NREL). Furthermore, encapsulated tandem devices based on Me-Ph2mPACz demonstrate exceptional stability, retaining 83% of their initial efficiency after 1000 h of maximum power point tracking under one-sun illumination at 85 °C in air. This work opens an avenue for designing high-performance and durable self-assembled molecules for perovskite tandem photovoltaics.

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Introduction

Solution-processed inverted perovskite solar cells (PSCs) represents a cost-effective and scalable photovoltaic technology,^[1-3] and their compatibility with tandem architectures further enhances their commercial potential.^[4,5] Certified efficiencies have reached 27.3% for single-junction and 34.85% for perovskite/silicon tandem cells.^[6] This progress is closely linked to advances in hole-transport layers,^[7-10] particularly the emergence of self-assembled molecules (SAMs) as replacements for conventional hole transporting materials such as PTAA and PEDOT:PSS in inverted PSCs.^[11,12] Composed of an anchoring group, a linker, and a conjugated terminal unit, SAMs bind to metal oxides and enable efficient hole extraction with low optical and electrical losses. Furthermore, their molecular design allows fine-tuning of energy levels, extraction capability, and stability, making SAMs adaptable to various bandgaps and device configuration. However, a common limitation of SAMs lies in their inherent tendency toward aggregation, leading to non-uniform films, disordered packing, and poor charge transport properties. These issues increase interfacial non-radiative recombination and hinder further improvements in the efficiency and stability of both single-junction and tandem solar cells.

SAM aggregation originates from their intrinsic amphiphilic character,^[13,14] which drives micelle formation in common processing solvents rather than discrete molecules. To address this, several strategies focus on manipulating intermolecular interaction of SAM molecules during deposition or assembly. The co-solvent approach uses additives such as DMF or DMSO to dissociating micelles, promoting denser SAM coverage.^[14,15] By employing auxiliary molecules that disrupt aggregation through non-covalent interactions,^[16-19] molecular doping is another way to enhance film homogeneity and promotes favorable perovskite crystallization at the interface.^[20,21] In co-SAM systems, a second SAM molecule is added for enhances packing,^[22] wettability,^[23,24] and energy-level alignment while dispersing phosphonic acid clusters.^[25,26] Alternatively, rational molecular design can intrinsically suppress aggregation—via steric hindrance or conformational distortion—while preserving efficient charge transport.^[27-30] A key consideration in designing such molecular structures is the balance of π - π interactions. While excessive π - π stacking lead to detrimental aggregation, moderated interactions enable densely packed, ordered multilayers that support efficient hole transportation.^[31-33] Thus, advancing SAM-based interfaces requires ingenious molecular design strategies that simultaneously inhibit aggregation and promote vertical charge conduction.

Herein, we synthesized a SAM, Me-Ph2mPACz, featuring a benzene-ring linker symmetrically functionalized with dual carbazole units at the *meta*-positions. Dimer interaction analysis reveals that this *meta*-disubstitution strategy enhances the continuity of intermolecular π - π stacking while reducing hydrogen-bond interaction between anchoring groups, thereby suppressing the formation of undesired large size SAMs clusters and yielding smaller molecular micelles. X-ray diffraction and grazing-incidence wide-angle X-ray scattering confirm that the strengthened π - π interactions promote a more uniform,

well-ordered, and compact multilayer configuration on the substrate. This optimized interface results in a wetSupplementary Table surface with more homogeneous potential distribution, improved energy-level alignment with the perovskite layer, and consequently more efficient hole extraction. Furthermore, it retards crystallization and alleviates residual strain during perovskite crystallization, reducing non-radiative recombination at the buried interface. As a result, wide-bandgap (1.68 eV) perovskite solar cells incorporating Me-Ph2*m*PACz achieve an increase in efficiency from 21.82% to 23.14%. The perovskite–silicon tandem devices achieved a champion PCE of 32.40% (certified 32.45% at NREL), with an improved V_{OC} 1.97 V, an FF of 83.1% and a J_{SC} of 20.40 mA cm⁻². Critically, the multilayer configuration of Me-Ph2*m*PACz on ITO/NiO_x exhibits robust thermal stability—with negligible variance after 1000 hours of continuous aging at 85 °C—which translates into exceptional device performance: encapsulated tandem devices retain 83% of their initial efficiency after 1000 h of MPPT under one-sun illumination at 85 °C in air. Our work demonstrates a rational molecular design strategy that concurrently suppresses SAM aggregation and enhances hole extraction, offering a promising approach to high-performance interfaces for perovskite photovoltaics.

Results

Fig. 1 illustrates the molecular design of the SAM, along with characterization of their self-assembly behavior. As shown in **Fig. 1a**, reference molecule Me-Ph*hp*PACz was previously reported for its good stability due to the rigid phenyl linker,^[34] and Me-Ph2*m*PACz was developed in his work. The latter was obtained by relocating the terminal 3,6-di-*tert*-butylcarbazolyl units to the two *meta*-positions relative to the phosphonic acid anchors. The synthetic routes are provided in **Supplementary Figure 1**, and the structures of both molecules were confirmed by ¹H, ¹³C, ³¹P NMR spectroscopy and high-resolution mass spectrum (**Supplementary Figure 2–17**). Frontier orbital analysis (**Supplementary Figure 18**) indicates more delocalized electron distribution in Me-Ph2*m*PACz, suggesting enhanced molecular stability.^[34–36] Density-functional theory calculations reveal a higher adsorption energy of Me-Ph2*m*PACz on NiO_x (3.41 eV) compared to Me-Ph*hp*PACz (2.46 eV), consistent with its larger interfacial charge transfer (**Fig. 1b**). The *meta*-dual-substitution also confers a larger dipole moment (1.27 Debye vs. 1.18 Debye for Me-Ph*hp*PACz) (**Fig. 1c**), which is expected to promote stronger anchoring and more effective energy-level modification on NiO_x.

To understand the self-assembly behavior, we analyzed intermolecular interactions within SAM dimers (**Fig. 1d** and **Fig. 1e**). In Me-Ph*hp*PACz, the substantial torsion angle between the carbazole and the phenyl linker disrupts π – π orbital overlap, leading to discontinuous stacking near the linker-terminal group junction. In contrast, the more planar *meta*-substituted molecular geometry of Me-Ph2*m*PACz with two large carbazolyl ring enables extended, continuous π – π interactions throughout the entire dimer, as visualized by the broader and denser contours of electron-density (red frame, **Fig. 1d**). This continuous π -electron system also weakens hydrogen-bonding between phosphonic acid anchors (**Supplementary**

Figure 19), reducing the bonding that typically hinders micelle disassembly during film formation. Additionally, **Supplementary Figure 20** indicates stronger charge transfer between Me-Ph2mPACz and the perovskite layer, irrespective of whether the SAM adsorbs in a parallel or perpendicular orientation on the perovskite surface. Dynamic light scattering (DLS) validates this trend: Me-PhpPACz contain polydisperse aggregates spanning hundreds of nanometers, whereas Me-Ph2mPACz yields clusters below 10 nm, reflecting a lower energy barrier for achieving uniform thin films (**Fig. 1f**). The degree of order in self-assembled films was probed by X-ray diffraction (XRD) on ITO/NiO_x substrates (**Fig. 1g** and **Supplementary Figure 21**). While Me-PhpPACz films show no crystalline peaks, a distinct reflection at $\sim 8^\circ$ appears for Me-Ph2mPACz, indicating a more ordered packing of SAM molecules. To gain a deeper understanding of the packing modes of the two SAMs, we characterized them using GIWAXS (**Fig. 1h**). Compared to Me-PhpPACz, the Me-Ph2mPACz sample exhibits weaker background signals (**Supplementary Figure 21**), suggesting a higher molecular coverage. Both SAMs display strong signals along the q_z direction, indicating a predominantly out-of-plane orientation. However, Me-PhpPACz shows multiple diffuse arcs in the out-of-plane direction at $q_z \approx 0.23, 0.45, 0.5, 0.7$, and $0.95 \text{ \AA}^{-1}$, respectively, indicating its less ordered packing. In contrast, Me-Ph2mPACz exhibits a sharp diffraction spot at $q_z \approx 0.3 \text{ \AA}^{-1}$, signifying ordered out-of-plane (OOP) π - π stacking. In the in-plane (IP) direction, both SAMs show weak π - π stacking signals, but for Me-Ph2mPACz, this signal shifts to a higher q value ($\approx 0.32 \text{ \AA}^{-1}$) and appears tilted with respect to the substrate. Considering that π - π packing is governed by the bulky carbazole groups, we attribute the weak IP π - π stacking signals to the layer that directly anchored to the ITO or NiO_x substrate via chemical bonds (the anchoring layer), while the out-of-plane signals originate from the stacking layer on the top of the anchoring layer, with both layers coexisting.^[13,37] This also explains why the IP signal of Me-Ph2mPACz is not perfectly parallel to the substrate: the *meta*-substitution of the carbazole groups in Me-Ph2mPACz induces a tilt in the π - π stacking direction of the anchoring layer. The above layer dominates the signal and is therefore expected to play a decisive role in determining the properties of the self-assembled film. From this perspective, the higher degree of order in Me-Ph2mPACz should be more favorable for the subsequent deposition of high-quality perovskite films and interface charge transfer, as will be discussed below.

We next correlated the properties of the self-assembled SAM films with their influence on the perovskite layer deposited above. Cyclic voltammetry (CV) measurements determined the areal molecular density of Me-Ph2mPACz to be $3.06 \times 10^{13} \text{ molecules cm}^{-2}$, which is 53% higher than that of Me-PhpPACz ($2.00 \times 10^{13} \text{ molecules cm}^{-2}$)^[22,38] (**Fig. 2a, Supplementary Figure 22**). The higher areal molecular density of Me-Ph2mPACz corresponds to better substrate coverage, owing to stronger SAM anchoring and easier micelle dissociation. Atomic force microscopy (AFM) was further used to measure the film morphology, and Me-Ph2mPACz showed a lower root-mean-square (Rq) roughness (4.85 nm) than Me-PhpPACz (4.41 nm) when deposited on ITO/NiO_x/ substrate (**Supplementary Figure 23**), indicating a more uniform surface. The enhanced uniformity was further corroborated electrically by Kelvin probe force microscopy

(KPFM). The potential root-mean-square of Me-Ph2mPACz (42 mV) was markedly lower than that of Me-PhpPACz (63 mV) (**Fig. 2b**), indicating a more homogeneous surface potential distribution.^[29] The ITO/NiO_x/Me-Ph2mPACz surface also exhibited a reduced contact angle (88.9 °) compared to ITO/NiO_x/Me-PhpPACz (74.2 °) (**Supplementary Figure 24**), which is more favorable for subsequent perovskite deposition. Reduced contact angle and higher molecular density suggest that there might be more exposed phosphonic groups of ITO/NiO_x/Me-Ph2mPACz surface, which helps retard the crystallization process and passivate the buried defects. The ordered and densely packed SAMs are expected to improve charge transport properties. Indeed, conductance measurements based on hole-only devices (ITO/NiO_x/SAM/Ag) showed an increase in conductance values, from 0.167 S for Me-PhpPACz to 0.211 S for Me-Ph2mPACz (**Fig. 2c**).

The well-ordered SAM morphology further translated into enhanced perovskite growth. AFM and scanning electron microscopy (SEM) indicated a smoother perovskite surface on Me-Ph2mPACz, with *Rq* of roughness decreasing from 10.9 nm to 8.64 nm (**Supplementary Figure 25**). SEM further revealed fewer PbI₂ residues on the perovskite surface that based on Me-Ph2mPACz (**Supplementary Figure 26a, b**), a finding consistent with the weaker PbI₂ signal in XRD patterns (**Supplementary Figure 27a**). Inspection of the buried interface showed that Me-Ph2mPACz facilitated cleaner detachment, often carrying adhered NiO_x nanoparticles due to its stronger anchoring—an evidence of its robust interfacial adhesion (**Supplementary Figure 26c, d**). Cross-sectional SEM images further confirmed that perovskites grown on Me-Ph2mPACz exhibited smoother surfaces and fewer vertical grain boundaries (**Fig. 2d**). To quantify residual stress of the resulted perovskite film, surface Grazing-incidence X-ray diffraction (GIXRD) was employed for $\sin^2\psi$ analysis. The perovskite film on Me-PhpPACz exhibited a compressive stress (slope = -0.17), whereas the film on Me-Ph2mPACz showed nearly stress-free characteristics (**Fig. 2e**). GIXRD of buried interface also showed improved crystallization (**Supplementary Figure 28**) for perovskite grown on the Me-Ph2mPACz. To further elucidate the roles of SAMs in perovskite film formation, we perform *in-situ* PL spectroscopy to monitor crystallization dynamic during spin-coating and thermal annealing. During the spin coating process, the wet perovskite film on ITO/NiO_x/Me-Ph2mPACz exhibited a stronger photoluminescence (PL) signal,^[39,40] indicating a higher density of nuclei (**Supplementary Figure 29a and Fig 29b**), which may be ascribed to the enhanced wettability and more uniform SAM multilayer. During the annealing process, the initial PL quenching followed by enhancement of the PL signal is typically attributed to the dissolution and recrystallization of crystal grain during solvent evaporation (**Fig. 29c and Fig. 29c**).^[39] The decrease in PL intensity observed at the final stage of annealing corresponds to the formation of grain boundaries and the accumulation of thermal stress. For perovskite deposited on ITO/NiO_x/Me-Ph2mPACz, the PL quenching and decreasing rate is much slower than that on ITO/NiO_x/Me-PhpPACz, indicating a slower Ostwald ripening process.^[41] Moreover, the continuously higher PL intensity during the final stage on ITO/NiO_x/Me-Ph2mPACz suggests that the retarded recrystallization led to fewer defects and less stress.^[42] The delayed recrystallization process might be

caused by the coordination interaction between P=O group in Me-Ph2mPACz and perovskite component, as mentioned above. Integrating these observations with the earlier analysis regarding micelle in **Fig. 1f**, we propose the following mechanism, as depicted in **Fig. 2f**: The enhanced π - π interactions in Me-Ph2mPACz promote the formation of small size and uniform solution aggregates, which readily dissociate upon deposition to form a dense and ordered bilayer structure multilayer (~ 3.5 nm, **Supplementary Figure 23**) that comprising an anchored layer adjacent to the substrate and a face-on π -stacked overlayer. This configuration accelerates perovskite nucleation while delaying the crystallization process, thereby releasing residual stress of perovskite film during the crystal growth. In contrast, Me-PhpPACz forms larger aggregates with limited dissociation, yielding a less uniform multilayer that imparts residual stress to the overlying perovskite. The effective mitigation of such stress is crucial for reducing defect density and enhancing the long-term operational stability of perovskite solar cells.

We further monitor the durability of the two SAM molecules and the multilayer of Me-Ph2mPAC, since it is vital to the long-time operation of device. TGA analysis showed Me-Ph2mPACz (440 °C) exhibits a significantly higher 5% thermal decomposition temperature than Me-PhpPACz (316 °C), indicating its better thermal stability (**Supplementary Figure 30**). As shown in **Supplementary Figure 31**, the crystallization signal of the Me-Ph2mPACz film remained almost unchanged after thermal aging, indicating that the film still retained its original uniform and ordered multilayer conformation. This finding was further confirmed by AFM test results. **Supplementary Figure 32** illustrates the changes in the overall morphology of the films before and after thermal aging. For the aged Me-Ph2mPACz, only negligible changes in roughness and morphology were observed. However, the roughness distribution of the Me-PhpPACz film exhibited a significant deformation, with a noticeable increase in areas of higher roughness. This was reflected in the morphological images by the emergence of numerous white spots of varying sizes, which should be attributed to the desorption and agglomeration of Me-PhpPACz caused by long-time heating, due to its lower adsorption energy and weaker intermolecular interactions. Additionally, we rinsing the aged SAM films with DMF and compared the changes in SAM coverage by measuring the P/Sn elemental ratio before and after treatment using XPS (**Supplementary Figure 33** and **Supplementary Table 1**). As shown in **Supplementary Table 1**, the molecular retention of Me-Ph2mPACz self-assembled films after treatment was much higher than that of Me-PhpPACz, demonstrating that Me-Ph2mPACz possesses stronger resistance to thermal and solvent rinsing compared to Me-PhpPACz.

The dense, uniform, and ordered SAM structure is expected to facilitate efficient hole extraction. We therefore systematically investigated the carrier dynamics of perovskites deposited on Me-PhpPACz and Me-Ph2mPACz. Ultraviolet-visible absorption (UV-vis) of the 1.68 eV perovskites revealed that Me-Ph2mPACz does not alter the bandgap but enhances the absorption (**Supplementary Figure 34a**), which is likely due to improved film crystallinity (**Fig. 27a**). Steady-state photoluminescence (PL) showed a 3.4-fold increase in PL intensity on Me-Ph2mPACz (**Supplementary Figure 34b**), indicating suppressed

trap-assisted non-radiative recombination.^[43] Results of large-area PL mapping ($50 \times 50 \mu\text{m}^2$) further confirmed the stronger and more homogeneous PL emission from the Me-Ph2mPACz-based sample (**Supplementary Figure 35**). Consistently, its PLQY was also much higher and more spatially uniform (**Fig. 3a**), pointing to the role of an ordered SAM in suppressing nonradiative recombination.

Ultraviolet photoelectron spectroscopy (UPS) was used to determine the energy-level alignment (**Supplementary Figure 36**). The Fermi levels of ITO/NiO_x/Me-PhpPACz and ITO/NiO_x/Me-Ph2mPACz were measured at -4.52 eV and -4.40 eV , respectively (**Fig. 3b**), which is consistent with the contact-potential trend from KPFM (**Supplementary Table 2**). The ionization energies (IE) of the substrates were -5.41 eV and -5.37 eV , respectively aligning with the calculated HOMO levels of the respective SAMs. A vacuum level shift of 0.01 eV was observed between the perovskite and the Me-PhpPACz buried interface, which increased to 0.12 eV for the perovskite/Me-Ph2mPACz, respectively. The higher vacuum level shift in Me-Ph2mPACz/perovskite interface can be attributed to the out-of-plane dipole formed by its face-on, densely packed SAM structure.^[44] The ionization energy (IE) of the Me-Ph2mPACz film (IE = 5.37 eV) was closely matched to that of the perovskite film (IE = 5.38 eV), which contributed to the reduction of the hole transfer barrier and suppression of nonradiative recombination at the interface.^[33]

Time-resolved PL (TRPL) decay curves were fitted biexponentially (**Fig. 3c** and **Supplementary Table 3**). The perovskite on Me-Ph2mPACz exhibited a shorter τ_1 (17.34 ns vs. 21.48 ns for Me-PhpPACz), associated with faster interfacial charge extraction, and a longer τ_2 (133.14 ns vs. 117.04 ns), reflecting suppressed non-radiative recombination. The derived differential lifetimes of TRPL results are shown in **Fig. 3d**. The first interval in the differential lifetime profile of Me-Ph2mPACz is shorter than that of Me-PhpPACz. This process is dominated by the charge transfer process, verifying the superior interfacial charge extraction of Me-Ph2mPACz.^[45] These results confirm that the extended conjugation and enhanced π - π stacking in Me-Ph2mPACz promote hole extraction and reduce nonradiative recombination losses. Femtosecond transient absorption (fs-TA) spectroscopy further confirm this conclusion: a faster decay of the ground-state bleaching signal at $700\text{--}720 \text{ nm}$ was observed for Me-Ph2mPACz (**Fig. 3e**, **Supplementary Figure 37** and **Supplementary Table 4**), indicating accelerated hole transfer.^[46] Quasi-Fermi-level splitting (QFLS) was derived from the PLQY distributions (**Fig. 3f**). The Me-Ph2mPACz/perovskite sample showed a higher and more narrowly distributed QFLS (median = 1.318 eV), compared with the broader and lower distribution for Me-PhpPACz (median = 1.264 eV). The improvement is attributed to the uniform SAM coverage, ordered molecular orientation, and reduced perovskite lattice strain.

To evaluate the implied photovoltaic performance, intensity-dependent steady-state PL was used to construct *pseudo-J-V* curves (**Fig. 3g**). Although interfacial recombination lowers the simulated efficiency relative to the optical limit, the Me-Ph2mPACz stack consistently outperformed the Me-PhpPACz stack, achieving a pseudo-PCE of 24.58% with an implied V_{OC} of 1.307 V and a *pseudo*-FF of 86.12%

(**Supplementary Table 5**). Finally, electroluminescence external quantum efficiency (EQE_{EL}) measurements were performed to quantify non-radiative voltage losses in full devices (**Fig. 3h**). The Me-Ph2mPACz-based device exhibited a significantly higher EQE_{EL} (0.82% at J_{SC}) than the Me-PhpPACz-based device (0.15%). Correspondingly, the non-radiative V_{OC} loss was reduced from 168 mV for Me-PhpPACz to 124 mV for Me-Ph2mPACz (**Fig. 3i**), quantitatively confirming the superior interfacial passivation and charge-extraction capability of the designed SAM.

To explore the impact of the SAMs on device performance, we fabricated 1.68 eV wide-bandgap perovskite solar cells (WBG PSCs, active area: 0.0804 cm^2) with the architecture of Glass/ITO/ NiO_x /SAM/Perovskite/ EDAI_2 /PCBM/ $\text{Zr}(\text{acac})_4$ /Ag (**Fig. 4a**). Statistical analysis of the key photovoltaic parameters (**Fig. 4b** and **Supplementary Figure 38**) confirms that Me-Ph2mPACz-based devices consistently outperform their Me-PhpPACz counterparts, regardless of whether the perovskite layer is deposited by gas quenching or the anti-solvent method (**Supplementary Figure 41**), indicating the versatility of the SAM multilayer. The champion Me-Ph2mPACz-based device achieves a PCE of 23.14% (23.06) at reverse scan with a V_{OC} of 1.26, a J_{SC} of 21.54 mA cm^{-2} , an FF of 85.3% and negligible hysteresis, a clear improvement over the Me-PhpPACz-based champion device (a PCE of 21.82%, a V_{OC} of 1.236 V, a J_{SC} of 21.09 mA cm^{-2} , an FF of 83.72% and significant hysteresis) (**Fig. 4c**). The gains in V_{OC} and FF stem from the enhanced hole extraction and suppressed non-radiative recombination enabled by the ordered SAM structure. This is further supported by a lower ideality factor (1.49 vs. 1.74 for Me-PhpPACz, **Supplementary Figure 39a**) and reduced dark current density (**Supplementary Figure 39b**) in Me-Ph2mPACz devices, indicating diminished Shockley-Read-Hall recombination. Additionally, we measured the impedance spectrum (**Supplementary Figure 40**). The R_{rec} obtained from equivalent circuit fitting of the PSC based on Me-Ph2mPACz is significantly higher than that of the device based on Me-PhpPACz, indicating that non-radiative recombination in the device is effectively reduced. The improved J_{SC} is consistent with the higher conductance of the Me-Ph2mPACz film and is corroborated by external quantum efficiency measurements, which show a slightly higher integrated current (20.51 vs. 20.46 mA cm^{-2}) without a shift in bandgap (**Supplementary Figure 33c**).

We subsequently integrated the SAMs-based perovskite top cell with a silicon heterojunction bottom cell to fabricate monolithic tandem devices (active area: 1.05 cm^2 ; structure and cross-sectional SEM in **Fig. 4d** and **Supplementary Figure 42**, respectively). The box plots of the nine independent devices based on different SAMs provides a direct comparison (**Supplementary Figure 43**). The photovoltaic values averaged over nine tandem devices indicate a PCE of 33.12%, a V_{OC} 1.97 V, an FF of 82.7% and a short-circuit current density (J_{SC}) of 20.35 mA cm^{-2} with Me-Ph2mPACz, compared with a PCE of 31.76%, a V_{OC} 1.96 V, an FF of 79.9% and a short-circuit current density (J_{SC}) of 20.29 mA cm^{-2} with Me-PhpPACz. To more intuitively demonstrate the superiority of the SAM molecular design in this work, we also fabricated tandem devices using 2PACz (commonly used as a control) as the HTL for a direct comparison under the

same conditions.^[10] However, the 2PACz-based champion device only showed an inferior PCE of 30.84% at reverse scan (30.38% at forward scan) with stabilized efficiency of 30.40%, due to a fairly low V_{OC} of 1.90 V and an FF of 79.9% (Supplementary Figure 43) (comparable to reported values). With an improved V_{OC} 1.97 V and an FF of 83.1% the tandem achieves a champion PCE of 33.40% at reverse scan (33.32% at forward scan) with stabilized efficiency of 33.17%, compared with a reverse scan value of 32.35% (31.95% at forward scan) with stabilized 32.11% for the tandem with Me-Ph ρ PACz. This tandem configuration has been sent to independently certified as 32.45% by the National Renewable Energy Laboratory (NREL) (Supplementary Figure 44). The observed significant improvements in V_{oc} and FF are probably attributable to the high performance of single-junction PSCs, which can be linked to the accelerated hole extraction and effective suppression of non-radiative recombination at the optimized SAM/perovskite interface. The photogenerated J_{SC} values of 21.35 and 20.48 mA cm⁻² were obtained for the perovskite and silicon subcells in tandem device with Me-Ph2mPACz, respectively (Fig 4f).

The robust and ordered SAM layer also enhanced operational stability. Unencapsulated single-junction WBG PSCs based on Me-Ph2mPACz retained 95% of their initial efficiency after 1,000 h of maximum power point tracking (MPPT) under 1-sun LED illumination at 25 °C in N₂, whereas Me-Ph ρ PACz-based devices degraded to 88% of their initial PCE (Supplementary Figure 45). For encapsulated perovskite-silicon tandem solar cells (TSCs) (seen in Supplementary Figure 46), the device based on Me-Ph2mPACz retained 83% of the initial efficiency after 1,000 h of MPPT under one-sun illumination at 85 °C in the air, outperforming their Me-Ph ρ PACz-based counterparts (only 35% of retainment) (Fig. 4g).

Discussion

In this work, we report a SAM, Me-Ph2mPACz, engineered through *meta*-dual-dimethyl-carbazole substitution. This molecular design employs enhanced π - π interactions to mitigate SAM aggregation, yielding a dense, well-ordered, and thermally robust multilayer on the substrate. The resulting SAM/perovskite interface effectively mitigates residual strain, homogenizes the surface potential, and optimizes energy-level alignment, thereby facilitating efficient hole extraction and significantly reducing non-radiative recombination. As a result, 1.68 eV single-junction perovskite solar cells deliver a power conversion efficiency of 23.14%, and perovskite-silicon tandem cells attain 33.40% (certified at 32.45% by NREL). And the Encapsulated tandem devices retain their initial performance after 1,000 h under continuous 1-sun LED illumination at 85 °C in the air. This study establishes a paradigm for designing high-performance and durable self-assembled molecules for perovskite tandem photovoltaics.

Methods

Materials

Conductive Indium tin oxide (ITO) substrates (15 Ω sq⁻¹) with 20 mm $\times$ 20 mm dimension were purchased from

Hua Nan Xiang Cheng Technology Co., Ltd. NiO_x nanoparticles (NPs) were synthesized according to previous work.^[47] Self-assembled monolayer (SAM) of Me-Ph η PACz and Me-Ph2 η PACz were synthesized according to synthetic route depicted in Supplementary Figure 1. Aluminum oxide (Al₂O₃) NPs dispersing in isopropanol (IPA), *N,N*-Dimethylformamide (DMF) and Dimethylsulfoxide (DMSO) were purchased from Sigma-Aldrich. Dimethylammonium iodide (DMAI), cesium iodide (CsI), formamidine iodide (FAI) and Ethanediimine dihydroiodide (EDAI₂) were purchased from Greatcell Solar. Lead bromide (PbBr₂), and lead chloride (PbCl₂) were purchased from Advanced Election Technology Co., Ltd. Lead iodide (PbI₂) was purchased from Alfa Aesar. [6,6]-phenyl-C61-butyric acid methyl ester (PCBM) was purchased from Nano-C. Zirconium (IV) acetylacetonate (Zr(acac)₄) purchased from Shanghai Aladdin Biochemical Technology Co., Ltd.

Chemical reagents and solvents used for SAM molecules synthesis: 1-bromo-4-iodobenzene, 1-bromo-3,5-difluorobenzene, triethylamine, bromotrimethylsilane (TMSBr), extra-dry DMF, 1,4-dioxane, dichloromethane (DCM) methanol, deuterium reagents were purchased from J&K Scientific. 3,6-imethyl-9H-carbazole, Cesium carbonate (Cs₂CO₃), potassium acetate, palladium(II) acetate, 1,1-bis(diphenylphosphino)ferrocene (dppf), and diethyl phosphite were purchased from Energy Chemical Co. Ltd. Anhydrous sodium sulfate (Na₂SO₄) and toluene was purchased from Sinopharm Chemical Reagent Co., Ltd. Copper(I) iodide and cyclohexane-1,2-diamine (DMACH) was purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Potassium phosphate (K₃PO₄) was purchased from Shanghai Macklin Biochemical Co., Ltd.

Fabrication of single-junction WBG PSCs

The pre-patterned ITO substrates were cleaned and treated with UV-ozone for 20 min. The 10 mg mL⁻¹ NiO_x NPs in deionized water and IPA (3:1) were spun onto ITO substrates at 3000 r.p.m. for 30 s and annealed at 100 °C for 10 min under ambient conditions. Then, the obtained NiO_x films were immediately transferred into a nitrogen-filled glovebox. A 0.5 mg/mL solution of Me-Ph η PACz or Me-Ph2 η PACz (EtOH:DMF=3:1) was spin-coated statically at 5000 rpm for 30 s onto ITO substrates and annealed at 150 °C for 10 min. The perovskite absorber layers were deposited in the N₂-filled glove box with controlled H₂O and O₂ levels < 0.1 ppm, and the temperature was monitored to be 22–24 °C. The perovskite precursor with a composition of Cs_{0.3}FA_{0.6}DMA_{0.1}Pb(I_{0.87}Br_{0.13})₃ (1 M) was prepared by dissolving stoichiometric amounts of PbI₂, PbBr₂, PbCl₂, CsI, FAI, and DMAI in a mixed solvent of DMF and DMSO (4:1, v/v), with an additional 2 mol% PbCl₂ added to promote crystallization. 50 μ L of the precursor solution was drop-cast onto the hole-transport layer (HTL) substrate. The substrate was spun at 2000 r.p.m. for 15 s with an acceleration of 1000 r.p.m. s⁻¹ at first, and then at 4000 r.p.m. for the 45 s with an acceleration of 2000 r.p.m. s⁻¹, the nitrogen gun was vertically positioned 5 cm above the top of the substrates and the gas flow started after 30 s of the spin and duration of 20 s when the gas flow pressure was 0.3 MPa. Then, the perovskite wet-films were annealed at 100 °C for 20 min after aging 50s. For the anti-solvent method, the perovskite precursor concentration was increased to 1.15 M, and 200 μ L of anisole was dropped onto the film 10 s before the completion of spinning. The resulting perovskite film was immediately annealed at 100 °C for 20 min without the aging step. 1 mg mL⁻¹ Ethanediimine dihydroiodide (EDAI) was spun-coated dynamically at 5000 rpm for 30 s onto perovskite layer and annealed at 100 °C for 5 min. The 23 mg mL⁻¹ of PC₆₁BM solution in chlorobenzene (CB) was spin-coated at 2500 r.p.m. for 40 s and then annealed at 70 °C for 10 min. 2 mg mL⁻¹ of Zr(acac)₄ solution in IPA was spin-coated at 4000 r.p.m. for 40 s and without annealing. 100 nm of Ag were thermally evaporated onto the perovskite film at deposition rates of 0.1 Å s⁻¹-1 Å s⁻¹, respectively.

Fabrication of perovskite-silicon TSCs

The silicon bottom cells were fabricated using n-type Czochralski (CZ) silicon wafers with a resistivity of $1\text{--}5\ \Omega\cdot\text{cm}^{-1}$ and a thickness of approximately $250\text{--}300\ \mu\text{m}$. Initially, a double-sided symmetrical texturing process was then performed using a potassium hydroxide (KOH) solution, resulting in random pyramids with a height of $500\text{--}1000\ \text{nm}$ on both surfaces. Subsequently, the wafers underwent a standard RCA cleaning procedure and then loaded into a plasma-enhanced chemical vapor deposition (PECVD) cluster for the passivation and doped layers. Hydrogenated intrinsic amorphous silicon (i-a-Si:H) layers, each with a thickness of about $10\ \text{nm}$, were deposited on both the front and rear surfaces. This was followed by the deposition of a 20-nm -thick n-type hydrogenated nanocrystalline silicon (n-nc-Si:H) layer and a 30-nm -thick p-nc-Si:H layer on the front and rear side of the wafer, respectively. Subsequently, a 150-nm -thick indium tin oxide (ITO) layer was sputtered onto the rear side through a shadow mask defining an active area of $1.2 \times 1.2\ \text{cm}^2$. Silver (Ag) rear electrodes were then screen-printed onto the ITO layer, also through the same mask pattern. Finally, a 10-nm -thick ITO layer, serving as the tunnel recombination layer, was deposited on the front side. The completed silicon bottom cells were laser-cut into $2.5 \times 2.5\ \text{cm}^2$ substrates for subsequent tandem cell fabrication.

The silicon bottom cells were treated with UV–ozone for $20\ \text{min}$. A NiO_x NPs aqueous dispersion ($10\ \text{mg mL}^{-1}$) was deposited on the substrates by spin coating at $3000\ \text{rpm}$ for $30\ \text{s}$ with an acceleration of $1000\ \text{rpm/s}$, followed by thermal annealing at $100\ ^\circ\text{C}$ for $10\ \text{min}$. The substrates were then immediately transferred into a nitrogen-filled glovebox. Next, $1\ \text{mg mL}^{-1}$ SAMs solution in EtOH: DMF (v/v 3:1) was spin coated at $3000\ \text{rpm}$ for $30\ \text{s}$ with an acceleration of $1000\ \text{rpm/s}$, and annealed at $150\ ^\circ\text{C}$ for $10\ \text{min}$. Then, $100\ \mu\text{L}$ of the perovskite precursor solution ($1.7\ \text{M}$) was spin coated onto the substrates at $600\ \text{rpm}$ for $6\ \text{s}$ with an acceleration of $200\ \text{rpm/s}$, followed by $2,000\ \text{rpm}$ for $50\ \text{s}$ with an acceleration of $2000\ \text{rpm/s}$, and $6000\ \text{rpm}$ for $15\ \text{s}$ with acceleration of $2000\ \text{rpm/s}$. During the spin-coating process, $300\ \mu\text{L}$ of anisole was dropped onto the film at the $59\text{--}60\ \text{s}$. The resulting perovskite film was immediately annealed at $100\ ^\circ\text{C}$ for $20\ \text{min}$. Subsequently, a $1\ \text{mg mL}^{-1}$ of EDAI_2 solution in isopropanol was spin coated onto the perovskite films at $5,000\ \text{rpm}$ for $30\ \text{s}$, followed by annealing at $100\ ^\circ\text{C}$ for $5\ \text{min}$. The completed substrates were transferred into a thermal evaporation chamber, where $9\ \text{nm}$ of C_{60} was deposited at a rate of $0.05\ \text{\AA s}^{-1}\text{--}0.1\ \text{\AA s}^{-1}$. Subsequently, $21\ \text{nm}$ SnO_x was grown in an ALD chamber at $80\ ^\circ\text{C}$. Finally, $400\ \text{nm}$ of Ag and $90\ \text{nm}$ MgF_x was deposited in a thermal evaporation chamber. The final devices with a structure of nin-bottom wafer/ NiO_x /SAM/Perovskite/ C_{60} / SnO_2 /Ag/ MgF_x had an active area of $1.21\ \text{cm}^2$, with a masked aperture area of $1.05\ \text{cm}^2$ for photovoltaic measurements.

Molecular Property Simulation

The theoretical calculations were performed using the Gaussian 16 suite of programs^[48]. The structures of different SAMS were optimized at the B3LYP-D3/def2-TZVP level of theory^[49–53]. The negative electrostatic potential (ESP) region (red color) implies the electrophilic property, while the positive ESP (blue color) represent the nucleophilic one. The noncovalent interactions were performed via Multiwfn 3.8 programs^[54,55].

Geometry Model Optimization and Charge Density Difference/Interaction

The independent gradient model (IGMH) based on Hirshfeld partition^[56] was conducted for SAMS top to investigate the weak interaction between SAMS and perovskite using CP2K 2025.1 package^[57]. In particular, DZVP-MOLOP-SR-GTH basis sets for geometry optimization and TZVP-MOLOPT-SR-GTH basis sets for static calculations, and the Brillouin zone integration was sampled using a Monkhorst mesh of $2 \times 2 \times 1$. And the

color-filled gradient isosurface maps intuitively exhibit the interaction area and corresponding strength. The differential charge density of SAMS top and bottom interface were carried out using the Vienna Ab initio Simulation Package (VASP)^[58] with the projector augmented wave (PAW) method^[59]. The exchange-functional was treated using the Perdew-Burke-Ernzerhof (PBE) functional^[60]. The energy cutoff for the plane wave basis expansion was tuned to be 400 eV. Partial occupancies of the Kohn-Sham orbitals were permitted by the Gaussian smearing method and a width of 0.2 eV. The Brillouin zone was sampled with Monkhorst mesh of $2 \times 2 \times 1$ for the optimization for all the structures. The self-consistent calculations apply a convergence energy threshold of 10^{-5} eV, and the force convergence was tuned to be $0.05 \text{ eV } \text{\AA}^{-1}$.

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Data Availability Statement

All the data supporting the findings of this study are available within this article and its Supplementary Information. Any additional information can be obtained from corresponding authors upon request. Source data are provided with this paper.

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Acknowledgements

This work was financially supported by the National Key R&D Program of China (grant number 2024YFE0216900), the National Natural Science Foundation of China (grant numbers 52302229, 62574053 and 52561160144), Basic Research Program of Jiangsu (grant number BK20250343), Shanghai Science and Technology Commission Program (25DZ3002001), the State Key Laboratory of Photovoltaic Science and Technology of China (No. 202401030301) and the Fudan University Nano Information Science Innovation Support Center Project, Ministry of Education of China. We thank the staff of beamlines BL02U2, BL17B1 and BL18U1 at SSRF for providing the beam time and User Experiment Assist System of SSRF for their help, GIWAXS setup is also supported by National Natural Science Foundation of China (Grant No. 12175298). The authors thank Prof. Guangzheng Zuo and Dr. Yuqian Liu for technical support in EIS measurements; Dr. Donghao Miao, Dr. Qianqian Wang, and Mr. Liucheng Gao for assistance with in-situ PL measurements; Dr. Hong Liu for SEM technical support; and Dr. Xinxin Lian for WBG perovskite recipe optimization and moral support.

Author contributions

H.Z. convinced the idea and supervised the project. M.L. synthesized the SAMs, fabricated and characterized the single junction devices and performed the stability tests of device. Z.L. and Y.M. fabricated and characterized the tandem devices. Q. H. performed the DFT simulation. Y.C. (Yifan Chen), Z.W. and D.J. were responsible for the morphology characterization and analysis of perovskite film. R.X., D.Y. (DanniYu), R.H. and Y.M. were responsible for the encapsulation of tandem devices. D.Y. (DongshenYan) provided assistance with synthesis, and Y.W. provided the synthesis laboratory. Y.Y. performed GIWAXS measurements and analysis guidance. X.Z., S.W. and F.X. provided guidance for the analysis of perovskite and SAM thin films. M.L. wrote the first draft, W.C., Y.C. (Yifeng Chen), H.Z., J.G., Y.W. and J.C. revised the manuscript. All authors contributed mentioned on the manuscript.

Competing interests

The Authors declare the following competing interests: Hong Zhang and Ming Luo filled a Chinese patent related to this work (CN202511027586.5). The other authors declare no competing financial interests. All authors declare no competing non-financial interests.

Fig. 1| Molecular design. **a**, Chemical structure of Me-PhpPACz and Me-Ph2mPACz. **b**, Schematic illustration of the charge density difference when Me-PhpPACz (left) and Me-Ph2mPACz (right) contact NiO_x, respectively; Blue isosurface signifies electron depletion, while yellow represents electron accumulation. **c**, Molecular dipole moment and adsorption energy of SAMs on NiO_x. **d**, Dimeric stacking structure and interaction region indicator (IRI) of Me-PhpPACz (top) and Me-Ph2mPACz (bottom). **e**, IRI analysis of π - π interactions in Me-PhpPACz (top) and Me-Ph2mPACz (bottom) dimers. **f**, Particle size distribution in solutions of Me-PhpPACz and Me-Ph2mPACz as determined by DLS. **g**, XRD of Me-PhpPACz and Me-Ph2mPACz films deposited on NiO_x/ITO substrates. **h**, 2D GIWAXS image of Me-PhpPACz (left) and Me-Ph2mPACz (right) deposited on NiO_x/ITO substrates transferred to q -space.

Fig. 2| Properties of SAMs and influence on perovskite. **a**, Areal density of SAMs on NiO_x/ITO substrate. **b**, KPFM images and surface potential distribution of Me-PhpPACz/NiO_x/ITO (left) and Me-Ph2mPACz/NiO_x/ITO (right). The scale bar is 2 μ m. **c**, Conductance of Me-PhpPACz/NiO_x/ITO and Me-Ph2mPACz /NiO_x/ITO measured in N₂-saturated ACN solution under different voltage scan rates. **d**, Cross-sectional SEM images of the perovskites on Me-PhpPACz/NiO_x/ITO (left) and Me-Ph2mPACz/NiO_x/ITO substrates (right). The scale bar is 500 nm. **e**, Linear fit of 2θ - $\sin^2\psi$ of the perovskite films on Me-PhpPACz/NiO_x/ITO and Me-Ph2mPACz/NiO_x/ITO substrates. **f**, Schematic diagram of perovskite growth on the (left, with lattice stress) Me-PhpPACz/NiO_x/ITO and (right, with mitigated lattice stress) Me-Ph2mPACz/NiO_x/ITO substrates.

Fig. 3| Carrier dynamics of perovskite. **a**, PLQY mapping of perovskite films on Me-PhpPACz /NiO_x/ITO and Me-Ph2mPACz/NiO_x/ITO. **b**, energy level diagrams of the perovskites and SAMs/ NiO_x/ITO; **c**, TRPL of perovskite film on Me-PhpPACz/NiO_x/ITO and Me-Ph2mPACz/NiO_x/ITO. **d**, Differential lifetime derived from TRPL measurements in **c**. **e**, Femtosecond transient absorption (fs-TA) test of perovskite film on Me-PhpPACz/NiO_x/ITO and Me-Ph2mPACz/NiO_x/ITO. **f**, QFLS calculated from **a**. **g**, *pseudo-JV* curves constructed from Me-PhpPACz- and Me-Ph2mPACz-based stacks. **h**, EQE_{EL} curves of the Me-PhpPACz-and Me-Ph2mPACz-based devices. **i**, Nonradiative voltage losses calculated from EQE_{JSC} in (**j**).

Fig. 4| Photovoltaic performance and stability. **a**, Schematic illustration of single junction 1.68 eV PSCs. **b**, Statistics of PCE and V_{OC} of single junction 1.68 eV PSCs. **c**, J - V characteristics of the champion single junction 1.68 eV PSCs. The box plots show the median (central line), 25th to 75th percentiles (box bounds), minima to maxima (whiskers) and individual data points (circles). **d**, Schematic illustration of two terminal (2T) perovskite/c-Si TSC. **e**, J - V characteristics of the champion TSCs. **f**, EQE curves of the Me-Ph2mPACz-based TSC. **g**, MPPT of encapsulated TSCs under one-sun LED illumination at 85 °C in the air.

Editorial summary:

High-performance perovskite tandem solar cells are limited by uneven, disordered self-assembled monolayer (SAM) acting as hole-selective layers. Luo et al. design a SAM that yields uniform, stable interfaces, reducing non-radiative losses and boosting efficiency and durability

Peer review information: *Nature Communications* thanks Jingjing Xue, Stefaan De Wolf, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.







